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UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

Bachelor of Science in Computer Science

First Year Examination - Semester 1 - 2020

SCS 1202 — Programming using C — (Part A)

(2 hours for both parts)

ANSWER ALL QUESTIONS

Pages = 8 **Questions = 2**

To be completed by the candidate

Index Number | | | | | | | | |

Important Instructions

- This paper is in two parts, **Part A (8 pages** and **Part B (8 pages)**. The candidate **must answer all questions**. The details are as follows:
  Part A: 50 marks
  Part B: 50 marks
- **Write your answers only in the answer boxes provided** on this question paper.
- The candidate must answer the two parts separately.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the examination hall by a candidate.
- Questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
- Please write your answers only in English.

To be completed by the examiners

1	
2	
Total	

1. (i) Select **the correct or the most suitable choice** for each of the *eight* questions below and write the letter of the selected choice in the answer box provided.

i. Which of the following statements are true?

A. The *C standard library* consists of compiled codes.

B. *printf* and *scanf* are examples of C standard library routines.

C. When a C program is compiled, it is *automatically linked* with the relevant C standard library routines.

(a) A and B only	(b) A and C only	(c) B and C only	(d) all A, B and C

ANSWER: $\square$

ii. Which of the following gives a correct listing of the *built-in* data types in the C language?

(a) array, float, integer, string	(b) array, char, int, string
(c) char, double, float, int	(d) char, float, integer, str

ANSWER: $\square$

iii. If n, x and y are integers and n is 6, then the C statements x=n++; y=++x; would result in

(a) n=6, x=6 and y=7.	(b) n=6, x=7 and y=7.
(c) n=7, x=6 and y=7.	(d) n=7, x=7 and y=7.

ANSWER: $\square$

iv. Which of the following statements are true?

A. The C *Pre-processor* processes all lines that begin with # in a C program.

B. `#include <stdio.h>` is an executable file that could be run.

C. `stdio.h` file is required to be included in a C program when input/output routines (e.g., printf) are used.

(a) A and B only	(b) A and C only	(c) B and C only	(d) all A, B and C

ANSWER: $\square$

v. Consider the following statement that is written in a C code:

```
#define func(A,B)   ((A)<(B) ? (A):(B))
```

Assume that in the same code, the following statements are written:

```
x = 9;
y = 15;
z = func(x,y);
```

Then, what will be placed in z?

(a) ?	(b) :	(c) 9	(d) 15

ANSWER: $\square$

vi. If *ch* is correctly defined as a character variable, which of the following is the syntactically correct C code fragment?

```
(a)
if (ch == 'A' || ch == 'E' || ch == 'I' || ch=='O' || ch == 'U')
    printf("%c is a vowel.\n", ch);
else
  printf("%c is not a vowel.\n", ch);

(b)
if (ch == 'A' || ch == 'E' || ch == 'I' || ch=='O' || ch == 'U');
    printf("%c is a vowel.\n", ch);
else;
  printf("%c is not a vowel.\n", ch);

(c)
if (ch == 'A' || ch == 'E' || ch == 'I' || ch=='O' || ch == 'U')
    printf(%c is a vowel.\n, ch);
else
  printf(%c is not a vowel.\n, ch);

(d)
if (ch == 'A' || ch == 'E' || ch == 'I' || ch=='O' || ch == 'U');
    printf("%c is a vowel.\n");
else;
  printf("%c is not a vowel.\n");
```

ANSWER: ☐

vii. Which of the following C constructs can be used for repeating the same task?

A. *for*

B. *switch*

C. *while*

(a) A and B only	(b) A and C only	(c) B and C only	(d) all A, B and C

ANSWER: ☐

viii. What would be the output of the following C code?

```
#include <stdio.h>
  main(){
    short I=12;
    short J=10;
    short N;
    N = I&J;
    printf("N=%d\n",N);
}
```

(a) 8	(b) 10	(c) 12	(d) 14

ANSWER: ☐

[8 marks]

(ii) Explain the translation and linking that take place when a C program that uses the *printf* function is compiled.

....

....

....

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....

....

....

....

[3 marks]

(iii) What is the problem associated with the following code? Explain how you can eliminate that problem.

```
#include <stdio.h>

int main () {
  int a = 10;

  do {
      printf("value of a: %d\n", a);
  }while( a < 20 );

  return 0;
}
```

....

....

....

....

....

....

[2 marks]

(iv) Using a suitable *for* construct, explain the use of *break* and *continue*.

....

....

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....

....

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[3 marks]

(v) Explain *local*, *external* and *extern* variables identifying the differences.

....

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....

....

....

[3 marks]

(vi) What will be the output of the following C code?

```
#include <stdio.h>

int main() {
  char  a;
  a = 'X';
  printf("a=%c\n",a);
  printf("a=%d\n",a);
}
```

Note: The decimal representation of character X in the ASCII code is 88.

....

....

[3 marks]

(vii) What will happen if `a = 'X'` in the above code in (vi) is changed to `a = 'XY'`?

....

....

[3 marks]

2. **(i)** **Fill in** the blanks indicated by the dashes (--------) to complete the following code fragment to find the whether an input number is *even* or *odd.*

```
int main() {
  int n;

  printf("Enter an integer\n");
  scanf("%d", &n);

  <------->

    <------->;

  <------->

    <------->;

   return 0;
}
```

[4 marks]

(ii) **Fill in** the blanks indicated by the dashes (--------) to complete the following code to find the *power* of a number using the technique of **recursion**.
Note: Given below is an output from a program run:

```
Enter base number: 3
Enter power number(positive integer): 4
3^4 = 81
```

```
#include <------->

int <-----------------------------------> ; /* function prototype */

main() {
```

```
    int base, a, result;
    printf("Enter base number: ");
    scanf("%d", &base);
    printf("Enter power number(positive integer): ");

    scanf(<----------->);

    result = power(<------------------->);

    printf(<-------------------->);
}

int power(int base, int a) {

    if (<------->)
        return (<----------------->);
    else
        return <------->;
}
```

[8 marks]

(iii) Multiplication between a matrix A and a vector x in which the number of columns in A equals the number of rows in x is defined as shown in the figure.

$$\mathrm{Ax} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}$$

Figure 1: Matrix - vector multiplication

A program written in C is needed to multiply an integer 8×8 matrix A with an integer vector x to get the result vector y.

An *incomplete* program made for the purpose is given below. It is to run in a continuous loop until EOF (Ctrl+D) is pressed. At each iteration, the A matrix and the x vector are input interactively by the user. After the computation, the result vector y should be displayed on the screen on a single line with tab spaces between elements.

Complete the code by **filling-in** the 13 blanks indicated by dashes (--------).

```
/* <---------------------------------------------> */

#include <stdio.h>
#include <stdlib.h>

#define N      <------->

int  A<------->;

int  x<------->;
```

```
int  y<------->;

int main(int argc, char *argv[])
{
  int i, j, choice;

  do {
    /* input matrix A  */

    for (<------------------------->)

      for (<-------------------------->) {
        printf("Enter A[%d][%d]\n",i,j);
        scanf("%d",&A[i][j]);
      }

    /* input vector x  */

    for (<-------------------------->) {

      printf("Enter x[%d]\n",i);

      scanf("%d",&x[i]);
    }

    /* do the multiplication */
    for (i=0; i<N; i++) {

      <------->

        for (j=0; j<N; j++)

          <-------------------------->;
    }

#ifdef PRINT
    printf("Answer y:\n");
    for (i=0; i<N; i++)

      printf(<-------------------------->);
    printf("\n");
#endif

    printf("Press ENTER key to continue and ^D to exit!\n");
    choice = getchar();
    choice = getchar();

  } while ( choice != <------->);

  return(0);
}
```

[13 marks]

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University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination — Semester I— 2020

SCS1202 — Programming using C - Part B

(Two (2) Hours)

Answer ALL questions

Number of Pages = 8 Number of Questions = 2

To be completed by the candidate								
Index Number								

Important Instructions to candidates:

- The medium of instruction and questions is **English**.
- Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Write your index number on each and every page of the question paper.
- The duration of the paper is **Two (2)** hours for **both parts A & B**.
- This paper, **part B** has **2** questions on **8** pages.
- Answer **all** the questions in this **part B**
- Each question carries exactly **25 marks**.
- **Write your answers on the space provided** on this question paper.
- Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.

To be completed by the examiners

1	
2	
Total	

Index Number | | | | | | | | |

1. (a). Only ONE answer is correct in the following 10 MCQs. Underline the **best suite asnwer** among the given options.

[2 x 10 marks]

i. Pointer is a variable type that can hold

A. decimal values.
B. the memory address of another variable.
C. the values of complex data structures.
D. any type of arrays.
E. FILE variables.

ii. What among the following is NOT true?

A. Pointers are more efficient in handling the data tables.
B. pointers increase the execution speed.
C. pointers use to passing data by value.
D. Pointer arrays give a convenient method for storing strings.
E. Pointers reduce the length and complexity of a program.

iii. Consider the following line of code written in C.

```
x = &y;
```

Which of the following is true regarding the variables x and y above?

A. both x and y are integers.
B. y is a pointer and x is a variable of same type as y.
C. x is getting the value stored in y's memory address.
D. data type of both x and y are same.
E. y is equal to the memory address of x.

iv. Consider the following signature of a function written in C.

```
void xyz(int* a) {}
```

Which of the following is true according to the function definition given above?

A. function *xyz* can take an integer array as the parameter.
B. *xyz* takes parameters as passing by value.
C. *xyz* returns an integer.
D. a is an integer variable.
E. *xyz("UCSC");* is a valid call of this function.

v. If *sizeof(x)* returns the value 4, the variable type of x is

A. char.
B. double.
C. float.
D. int.
E. struct.

Index Number | | | | | | | | |

vi. If *vpr* is double pointer variable and the value of *vpr* is 000A83E0, the value of *vpr++* is;

A. 000A83E1
B. 000A83E4
C. 000A83E8
D. 000A83EA
E. 000A83F0

vii. Consider following piece of code written in C.

```
int arr[5] = {12, 23, 56, 87, 94};
int *ptr = arr;
printf("\%d,␣", *ptr+2);
printf("\%d", *(ptr+2));
```

What would be the values printed in the screen when we run this program?

A. 56, 56
B. 23, 23
C. 14, 23
D. 14, 56
E. 56, 94

viii. What is true among the following regarding the *structures* (struct) in C.

A. struct is another simple variable type.
B. members of a struct should be in same type.
C. struct can have one or more members in it.
D. members of a struct can have the same name.
E. size of a struct is equal to the sum of the sizes of its member variables.

ix. Consider following piece of code written in C.

```
struct datatype_1 {int dd, mm, yy;};
struct datatype_2 {int dd, mm, yy;};
void main(){
struct datatype_1 d1 = {12, 09, 2020);
struct datatype_2 d2;
d2 = d1;
}
```

What is the expected outcome from the following when you run the above code.

A. d2.dd = 12
B. d2.dd = 12, d2.mm = 9, and d2, yy = 2020
C. d2 can use as d1
D. give a runtime error
E. give a compilation error

Index Number | | | | | | | | |

x. Consider following piece of code written in C.

```
enum abc {a=1, b=3, c, d=4};
printf("%d", c);
```

According to the above code, the output will be printed in the screen is;

A. 0
B. 2
C. 3
D. 4
E. 5

(b). Consider following line of code written in C.

```
char *studentName[10];
```

i. how many students' names can store in this array? ....

[1 marks]

ii. what can be the length of a single name? ....

[2 marks]

(c). Assume that all of your students' information are stored in an array of struct called *students*. Write down the code line to print the age of the 50th student of the array.
Assume that the name of the struct array is *std* and the member variable which store the age is *Age*.

....

[2 marks]

Index Number | | | | | | | |

2. (a). Explain how the *Enumeration* datatype differ from *Array* data structure in C.

[3 marks]

(b). i. Explain how the *Union* datatype differ from *Structures* in C.

[3 marks]

ii. what you can say about the size of a Union?

[2 marks]

Index Number | | | | | | | | |

(c). Consider following program written in C.

```
void main(){
FILE *fp = fopen("MyFile.txt", "w");
char *name = "Kamal Perera";
int age = 23;
double gpa = 3.4;
if(fp==NULL) printf("error in creating the file...");

fprintf(fp, "student %d name: %s\n", 1, name);
fprintf(fp, "student %d age: %d\n", 2, age);
fprintf(fp, "student %d gpa: %.1f", 1, gpa);

fclose(fp);
}
```

i. Assume that the initial content of the *MyFile.txt* is,

```
My name is Nimal Perera
```

What are the new contents of *MyFile.txt* after executing the above code?

[5 marks]

ii. What would be the new content of *MyFile.txt*, if you run the same program again, after changing the second line of the code as follows;

```
FILE *fp = fopen("MyFile.txt", "a");
```

[3 marks]

Index Number | | | | | | | | |

(d). Consider following program written in C.

```
void main() {
FILE *fp = fopen("Records.txt", "r");
char ch = getc(fp);
while(ch!=EOF) {
ch = getc(fp);
printf("%c", ch);
}
if(feof(fp))
printf("\nEnd␣of␣file␣reached!");
}//main()
```

Assume that the initial content of the *Records.txt* is,

```
CS Students
```

What is the output displayed in the console after executing the above code?

[3 marks]

(e). Write a program in C to copy the entire content of a file to another file.

[6 marks]

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UNIVERSITY OF COLOMBO, SRI LANKA

UCSC

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Academic Year 2019/2020 – First Year Examination – Semester I – 2020

SCS 1201 – Data Structures and Algorithms I

TWO (2) HOURS (For Part A & B)

PART A

To be completed by the candidate

Examination Index No: ____________

Important Instructions to candidates:

1. The medium of instruction and questions is **English**.
2. **Write your answers in English** using the space provided **in this question paper**
3. Note that questions appear on both sides of the paper. If a page or a part of the question paper is not printed, please inform the supervisor immediately.
4. Write your index number on each and every page of the question paper.
5. This paper has **04** questions across **Part A** and **Part B**
6. Students are required to answer both **Part A** and **Part B** in **two hours**.
7. Answer **ALL** questions. There are **03** questions in **Part A** and **01** question in **Part B** of the paper.
8. Part A paper has 3 questions and 12 pages.
9. **Part A** of the paper contains **60 marks (20 Marks for each question)** and **Part B** of the paper will total to **40 marks.**
10. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
11. **Non-Programmable** calculators are **allowed.**

For Examiner's use only	
Question No	**Marks**
1	
2	
3	
Total	

1.

(a) The following circular queue can accommodate a maximum of eight elements with the following data.

Q= ----, ----, P, Q, R, S, T, ---

Front=2, Back=6,

If one adds the following items to the queue one item at a time, what would be the queue content with Front and Back values of the queue on each occasion?

(i) U
(ii) V
(iii) W
(iv) X

(4 Marks)

(i)

(ii)

(iii)

(iv)

(b) Consider the following stack.

Stack1 [] = [1,2,3,4,5]

Write a pseudocode algorithm to delete the middle element (which is 3) from the stack.

Hint:

(i) You may use pop (), push () operations
(ii) Additional stack data structures can be used
(iii) Stack properties cannot be violated during the process of the deletion.

(6 Marks)

(c) Consider the following pseudocode algorithm.

```
stack=[]
for  i=0 to 30 #for loop staring from 0 and ending with 29
    if i%3== 0 or  i%7==0 then
        stack.append( i )   #push element i
    else if  i % 4 == 0  then
        stack.pop()   #pop the element
print(stack)
```

Write down the output of the above program?

(5 Marks)

(d) Convert the following infix expression to its equivalent postfix expression, showing the stack contents for each step of conversion:

*(A/(B-C) *D + E)*

(5 Marks)

Symbol	Stack	Expression

2)

(a) Given a singly linked list and a key value for deletion. Write a pseudocode algorithm or C code segment to delete the first occurrence node with the given key from the linked list.

Data Values(keys) of the Linked List: 1,2,3,4,5,2

Key for deletion :2

You may use Start, Ptr1 and Ptr2 as pointer variables, and initially, all are pointing to the first node in the linked list. (i.e. to Node 1).

State any assumptions that you used.

(5 Marks)

(b) Write down the appropriate pseudocode commands or any programming language commands to perform the followings.

(i) to insert a new node as the first node in the linked list.
(ii) to insert a new node as the last in the linked list.

Data value of the new node(***key***)=22

Initial pointer to the new node =newnode

Initial pointer to the linked list =start

You should use the initial linked list (given in 2(a) above) to perform each of the above operations.

(5x2=10 Marks)

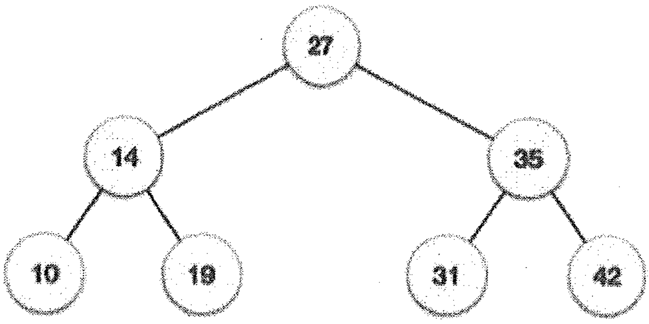
(i)   (ii)

(c) Write a pseudo algorithm or C code segment to find the total number of occurrences of the key-value 2 from the initial linked list (given in 2(a) above).

(5 Marks)

3.

(a) Consider the following binary search tree.



The following algorithm can be used to insert a new node to a binary search tree.

```
Input: binary search tree T, node v, element e //e and v are the value of the new node
and value of the current parent node respectively.
Output:
add(T, v, e){
        if(T.isLeaf(v)){
            if(v.element()>=e)
                    add element e as v's left child
            else
                    add element e as v's right child
        } else {
        if(v.element()>=e)
                    add(T, T.leftChild(v), e)
            else
                    add(T, T.rightChild(v), e)
  }
}
```

One wants to insert node number 28 to the above tree.

Explain how the above recursive algorithm works when inserting the node.

(4 Marks)

(b) One wants to delete the node number 27 form the above initial tree. Discuss how the node is deleted and what the replacing nodes are with respect to this deletion operation.

(4 Marks)

(c) What are the features to be satisfied when construing an AVL tree?

(2 Marks)

(d) Discuss the AVL tree property violation conditions during the construction of the AVL tree. You should use suitable diagram(s) to answer the question.

(4 Marks)

(e) Create the AVL tree using the following set of integer values. Discuss if any violations are indicated in the AVL tree construction. If there are violations, show how to eliminate them during the construction of the AVL tree.

Integer Values: 12,8,4,14,13,10,9

(6 Marks)

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BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Academic Year 2018/2019 – First Year Examination – Semester I – 2020

SCS 1201 – Data Structures and Algorithms I

TWO (2) HOURS (For Part A & B)

PART B

To be completed by the candidate

Examination Index No: ....

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6. Part B paper has **01** questions and **05** pages.
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8. Answer **ALL** questions. There are **03** questions in **Part A** and **01** Question in **Part B** of the paper.
9. **Part A** of the paper will total to **60** marks and **Part B** of the paper will total to **40** marks.
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Question No	**Marks**
1	
Total	

Index No: ....

Part B

Question 1

a) Consider an array (A) of n numbers in which a single peak needs to be discovered and printed on the output console. A peak in an array is defined as an element that is greater than both its neighboring elements.

Write a pseudo code to find a single peak as mentioned in the above scenario. Your answer should also consider and handle boundary cases such as the first and last elements of the array where a one neighbor is not available. Clearly state your assumptions if any.

[10 marks]

Index No: ....

b) Develop a cost function *f(n)* for the pseudocode by assigning a cost value for each line in part (a) above. Your answer should clearly state why such a cost is assigned to each line in the above code.

[05 marks]

c) Using the cost function *f(n)* developed in **part (b)** above, find the best case, worst case, and average case running time complexity for the above pseudocode. Provide an example array representation for each case above.

[10 marks]

d) Using suitable examples, compute the running time complexity of deleting an element using indexes from an array data structure. Clearly state your assumptions.

[05 marks]

Index No: ....

e) Compute the running time complexity and the space complexity for the pseudocode given below.

```
int a = 0, b = 0;
for (i = 0; i < N; i++) {
    a = a + rand();
}
for (j = 0; j < M; j++) {
    b = b + rand();
}
```

[10 marks]

****